GlucoTwin — system architecture

A patient-specific type 2 diabetes digital twin. Four input streams are harmonised onto one causal timeline, fused by the model layer, calibrated into prediction intervals and adverse-event probabilities, and served to two clinician interfaces. Raw patient records never leave the clinic.

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1 · SOURCES

Inside the clinic — raw records never leave

STATIC EHR

Health record

Real: age, sex, BMI, HbA1c, fasting glucose, lipids, medications. Simulated cohort adds diagnoses, eGFR, genetic markers.

REAL · OPEN

CGM · 145 patients

CGMacros (45 adults, PhysioNet) and Shanghai T2DM (100 patients, figshare). Removing this stream hurts the forecast most of all.

REAL + SIMULATED

Wearable

Fitbit heart rate and activity; sleep, HRV and steps simulated. At +60 min: **+0.04 mg/dL, CI includes zero** — reported, not hidden.

REAL + PRESETS

Meals

Logged macronutrients, plus 16 Indian presets from IFCT-2017. Meals are the **largest single gain**.

2 · HARMONISE & FUSE

One causal timeline

Common grid. Every cohort resampled to a single 5-minute schema with a shared column set; loader checksums verified.

Causal features only. No feature may use information from after the prediction instant — enforced in code and covered by tests.

Patient-grouped splits. Training, inner-validation, calibration and holdout patients never overlap, so no patient appears on both sides of an evaluation.

Gap and quality handling. Sensor gaps flagged rather than silently interpolated; a per-window quality score travels with the forecast.

Served feature set. Recent CGM history and its slope · time of day · meal macronutrients · post-meal activity · HbA1c and fasting glucose.

`src/data · src/features`

3 · MODEL LAYER

Fusion forecaster

Tuned LightGBM on the *change* from the current value at +30, +60 and +120 min. The served model.

Mechanistic twin

A per-patient glucose–insulin ODE. RK4 integration, extended Kalman filter state. Runs live in the console; sizes what-if

Hybrid physics + ML

Twin state added to LightGBM: 0.55 mg/dL lower error at +60 min. A GRU trained with an ODE-consistency loss.

Personalisation

Fine-tuning or recursive least squares on the patient's own days: about 1 mg/dL better after 3 to 5 days.

Federated training

FedAvg and FedProx across simulated clinics. Five-clinic and two-site layouts, plus a privacy noise sweep.

4 · CALIBRATE

80% intervals

Split-conformal and CV+. Served: 80.5% at +60 min; group-wise calibration holds every group near 80% (D-076).

Event probabilities

Platt and isotonic calibration for crossing 180, 140 and 70 mg/dL. Calibrated means 40% happens about 40% of the time.

Robustness & quality

Faults injected, causal features recomputed, scored on the same anchors. Quality score: completeness, recency, plausibility, rate of change.

Alerts & explainability

Alert-burden curves let a clinic trade sensitivity against false alarms explicitly. SHAP attribution over the fused feature set explains each forecast.

5 · CLINICAL OUTPUTS

Forecast + band

Trajectory to +120 min with its 80% interval and one plain sentence. The band widens with the horizon — that widening is the content.

Adverse-event risk

Calibrated chance of hyperglycaemia or hypoglycaemia inside the window, at the threshold the clinic chose. Thresholds follow the 2019 consensus.

What-if answers

Counterfactual meal size, carbohydrate, timing, an Indian preset swap, and a post-meal walk — through the monotone model and the twin.

Trust surface

Sensor data quality, personalisation depth, input tier available, feature attribution, and the limits printed beside each answer.

6 · INTERFACES

Web console `web/`

A single-page clinician console. Patient list with triage ordering, then per patient: forecast, what-if lab, glucose report (AGP, time in range, GMI per the 2019 consensus), health record, sleep and activity context, visit summary, and an evidence section carrying every cohort-level result.

Role views — clinician and care coordinator see different fields — with a session access log. Dark and light themes, mg/dL ⇄ mmol/L, readable at tablet width. The demo runs on 120 **simulated** patients with a three-day local calibration.

Streamlit application `make app`

Ten pages: Overview, Live twin, Glucose report, What-if lab, Twin state, Personalization, Input tiers, Data quality, Evidence, and a PDF export for the patient file.

Runs locally; no data leaves the machine. A units switch shows mg/dL or mmol/L, and the Evidence page reproduces every cohort-level table and figure.

Both interfaces read the same bundle, so they never disagree on a number. Every page carries the decision-support disclaimer: the system never recommends a dose or a drug.

METHOD & EVALUATION

- **Pre-registered.** Protocol, candidates and decision rule written to `docs/DECISIONS.md` before the run: 72 of the 76 entries.
- **Patient-grouped CV** with repeats, an inner validation split for every choice, and a reserved holdout the runner refuses to reuse.
- **Baselines it must beat:** persistence, per-patient autoregression, ridge, a GRU.
- **Paired patient bootstrap** intervals — over patients, never over time windows.
- **Also scored:** coverage, event AUROC, alert burden, plausibility, 87 matched breakfasts, input-tier ablations; stressed by transfer, faults and a synthetic Indian cohort. **Nulls published** at full size.

AI / ML MODELS AND TRAINING

- **Inputs → target.** Causal features per anchor: CGM lags, rolling statistics, least-squares slope and acceleration, time-of-day sin/cos, meal macros, post-meal activity, HbA1c, fasting glucose (heart rate tested, not served). Target is the *change* from the current value.
- **Served forecaster.** LightGBM, one head per horizon, trained with an absolute-error loss to match the metric it is scored on. 16 candidates, best three averaged, on inner-validation patients only.
- **Sequence model.** PyTorch GRU over the causal window with FiLM static conditioning; also the backbone of the physics-informed hybrid.
- **Constrained what-if.** A second LightGBM with monotonicity constraints: less carbohydrate can never predict a higher peak, by construction.
- **Uncertainty.** Split-conformal and CV+ intervals; Platt and isotonic probability calibration.

SERVED MODEL · MAE CGMACROS, AUROC SHANGHAI

METRIC	VALUE
MAE +30 min	**10.88**
MAE +60 min	**15.70**
MAE +120 min	**19.96**
AUROC >180, +60	**0.89**
AUROC <70, +60	**0.91**
80 % coverage	**80.5%**
Transfer gap	**3.4**

PRIVACY BY DESIGN

- Anonymised open or synthetic data only; no real patient record is in the repository.
- **Federated:** FedAvg matches pooled training (difference 0.09–0.12 mg/dL, intervals include zero) and beats a lone small clinic by ≈2 mg/dL.
- Data minimisation by design: only model updates are exchanged. No formal differential-privacy guarantee is claimed at this clinic count.
- Role views with a session log (display, not authentication); mapped to India's **DPDP Act** in `docs/limitations.md`.

ENGINEERING & SCOPE

- Python 3.11 · LightGBM · PyTorch · SHAP · Streamlit. Runs on a laptop CPU; no GPU required.
- One `make` target per stage; everything regenerates from source. CI runs ruff, black, pytest and a smoke run. Docker image, datasheet and model card included.
- One condition: **type 2 diabetes**. Decision support only — not a medical device.
- India, sleep and federated results are **simulations**: they test design, not clinical performance (sleep: at most 0.09 mg/dL over **9,000 patient-days**). Not prospectively validated.

Code MIT. Datasets keep their own licences. Simulated streams are labelled SIMULATED throughout.